Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building secure backend systems, AWS data pipelines, and high-quality testable services, with strong written documentation habits and a focus on shipping reliable components in fast-moving environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built production backend infrastructure for a multi-tenant AI platform, processing **1K+ AI jobs** and **5K+ monthly integration jobs** with **99%+ reliability** across SQL-backed pipelines and cloud storage.
- Wrote release-ready workflow logic, validation checks, and technical implementation details that reduced dashboard latency from **1.2s to 350ms** and improved recurring data-pipeline latency by **35%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered AWS-based ETL and API workflows using **Glue**, **S3**, SQL, and CI/CD automation, improving response times by **30%** and increasing pipeline efficiency by **68%**.
- Built and maintained automated unit and integration test flows across Windows, Linux, and macOS, reducing technical debt by **20%** and supporting production-grade release quality.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented **Python/Flask** services and concurrent processing workflows for an AI grading platform handling **10K+ requests/day**, with secure ingestion, background execution, and operational monitoring.
- Produced model-usage tracking, cost metrics, and AWS RDS-backed workflows for **150+ instructors**, reducing feedback turnaround time by **60%** while maintaining stable APIs under load.

Pandit Deendayal Energy University **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Built a malicious-URL detection pipeline over **650K+ URLs** with **21 behavioral indicators**, combining statistical feature engineering with ML and deep-learning evaluation.
- Achieved **96%+ detection accuracy** and generated measurable classification insights through systematic validation, error analysis, and model-comparison workflows.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built large-scale simulation and training pipelines over **100K+ trajectory logs** and **10K autonomous-driving episodes**, translating telemetry-style inputs into evaluation-ready system signals.
- Improved long-horizon planning by **18%** and closed-loop stability by **15%** through metrics-driven experimentation on autonomous-system behaviors.

Skills

Languages: Python, **Golang**, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: Flask, FastAPI, Django, Node.js, Express.js, REST APIs
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, **Parquet**, **Iceberg**
Cloud & Infrastructure: AWS (**EC2, S3, RDS, Lambda, ECS, Step Functions**), Microsoft Azure, Docker, Kubernetes, Terraform
AI/ML: PyTorch, TensorFlow, Transformers, NLP, LLM Systems, Feature Engineering, Hyperparameter Tuning
Engineering Practices: CI/CD, Distributed Systems, ETL Pipelines, Performance Optimization, Agile, System Design, Unit Testing, Integration Testing, Technical Documentation

Projects

Chat2SQL Repo | *Java, TypeScript, Docker, SQL*

- Extended the Chat2DB project into an AI-powered SQL workspace supporting **10+ databases**, cross-platform deployment, and structured data-processing workflows across Java, TypeScript, Maven, and Docker services.

Big-Brain Live | *FAISS, Pinecone, PostgreSQL, GCP*

- Engineered a cloud-based ingestion and indexing pipeline over **50K+ documents**, reducing retrieval latency by **40%** and supporting measurable retrieval-quality improvements through optimized data flows.

CaptionCraft - Image Caption Generator Live | *Transformers, PyTorch, Flask*

- Benchmarked captioning models on a **10K image / 100K caption dataset**, improving accuracy by **15%** (**81% to 96%**) while building inference APIs for scalable evaluation.